DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 11

Analytics engineer

Profession 11 of 10 in Development and IT.

What the profession means

An analytics engineer models data for analysis: transforms raw tables into clean, tested models that analysts and dashboards use, often with SQL and a transformation tool.

What professionals do every day

Writes transformation models, documents them, adds tests (for example uniqueness and not-null checks), and reviews changes with analysts.

Why businesses hire them, and what problems they solve

Analysts spend less time cleaning when the models are tested and documented once.

- Models that nobody documents.
- Silent duplicates after a join.
- Dashboards built on raw tables that change.

Skills you need

- SQL modelling.
- Testing and documentation.
- Version control.
- Understanding of business definitions.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
dbt Core	Transformations and tests (open source)	Free	Paid cloud	www.getdbt.com
SQLite	Learn SQL on a local file	Free	Free	www.sqlite.org
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Write one model that joins two tables.
Intermediate	Add a uniqueness test on the key.
Advanced	Document one model in plain words.

Projects by level

- **Beginner:** Clean model for a shop's orders.
- **Intermediate:** Customer model for a firm.
- **Advanced:** Revenue model shared by three dashboards.

Client-ready portfolio project

A tested model for a fictional shop: a clean orders table with a key test, a not-null test and a documented revenue definition.

Worked example

Worked example. An orders model joins customers once; a uniqueness test on the order id catches any join that creates duplicates.

Assignment with full solution

ASSIGNMENT

Uniqueness test on a joined model

Business problem: A join has started duplicating orders.

Steps

1. Write the model that joins orders to customers.
2. Count rows before and after the join.
3. Add a test that the order id is unique.
4. Record the result.

Expected output: The model, the row counts and the passing test.

SOLUTION (ANSWER KEY)

SQL

```
-- row counts: orders must not grow after the join
SELECT COUNT(*) FROM orders;
SELECT COUNT(*) FROM orders o JOIN customers c ON c.id = o.customer_id;
-- both return 9 for the sample data, so the join adds no duplicates
```

Common mistakes

- Trusting a join without counting rows.
- Undocumented business definitions.
- Changing models without tests.
- Naming columns differently in each model.

Freelancing opportunities

- Modelling projects for analytics teams.
- Model maintenance and tests.
- Briefs for 'dbt', 'analytics engineering', 'data modelling'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- dbt documentation

Part of **Master Freelancing: Data Analytics and Data Science**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.